Equilibrium and missing forces

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find a missing force when acceleration is zero.

SUCCESS CRITERIA - I can...

1. Recognize rest and constant velocity as possible equilibrium.
2. Find missing forces from a zero net-force condition.
3. Explain why turning is not equilibrium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 kg box slides left at 5 m/s on a rough horizontal floor after release. Draw forces and explain what follows about vertical and horizontal acceleration.

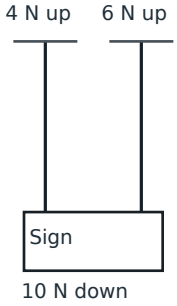
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Sigma F_x = 0$; $\Sigma F_y = 0 \Rightarrow a = 0$.
Equilibrium allows rest or constant velocity. Resolve missing forces along each axis. Use g = 9.8 N/kg.

Worked reasoning: A sign has weight 10 N down and two vertical upward pulls of 4 and 6 N. The vector sum is zero, so equilibrium is possible.

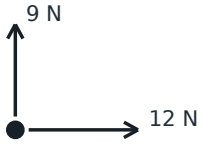
Your turn: Draw and label all three forces. Identify the equation that checks vertical balance.



Marzano check - circle: ① ② ③

03 Two component balance

A stationary object has 12 N right and 9 N up. A third force balances both. Find its components, magnitude and direction.

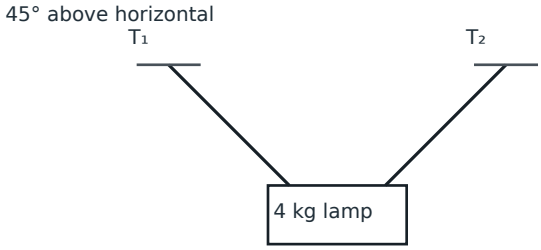


The dot represents the object. Third force not shown

Marzano check - circle: ① ② ③

04 Two-rope hanger

A stationary 4 kg lamp hangs from two identical ropes, each 45° above horizontal. Use g = 9.8 N/kg. Resolve the tensions into components and find each tension. Explain why each tension is larger than half the weight.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Change angle

A 10 kg sign hangs from two identical ropes. Compare ropes 30° above horizontal with ropes 60° above horizontal. Use g = 9.8 N/kg. Sketch both arrangements, calculate each tension, and explain which puts more tension in the ropes.

First arrangement: 30°

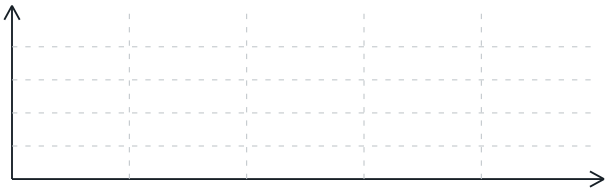


Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A=(7, -5) m and B=(-5, 9) m. Find R, magnitude and direction. Choose C so A+B+C=0 and sketch the route. Draw the component vectors and resultant; label lengths.

North / +y (m)



East / +x (m)

Marzano check - circle: 1 2 3

07 Exit check

A 3 kg lamp hangs at rest from a vertical rope. Use g = 9.8 N/kg. Find tension. Then redraw the forces if the lamp accelerates upward. Write the equation showing why tension must change.



Add and label force arrows.

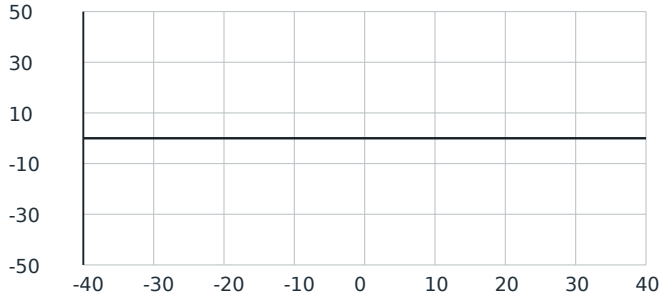
Choose the lamp as the object.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A stationary junction has a 30 N horizontal force right, a 40 N vertical force down, and one unknown force. Find the unknown vector, magnitude and direction.

Vertical force (N)



Horizontal force (N)

Marzano check - circle: 1 2 3